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GeoCBI: A modified version of the Composite Burn Index for the initial assessment of the short-term burn severity from remotely sensed data

Angela De Santis*, Emilio Chuvieco

Department of Geography, University of Alcalá, Calle Colegios 2, 28801 Alcalá de Henares, Madrid, Spain

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ABSTRACT

Burn severity estimation is a key factor in the post-fire management. Previous studies using remotely sensed data to retrieve burn severity, as measured by the Composite Burn Index (CBI), have found inconsistencies, since spectral indices work well in some ecosystems but not in others. These inconsistencies may be caused by the lack of spectral uniqueness in the CBI definition, or by the performance of the spectral indices used. This paper analyses the former aspect, using a simulation analysis to study the relationships between the CBI and reflectance. Subsequently, a modified version of this index, called GeoCBI, is proposed to improve the retrieval of burn severity from remotely sensed data. GeoCBI takes into account the fraction of cover (FCOV) of the different vegetation strata used to compute the CBI. Moreover, it also includes the changes in the leaf area index (LAI) for the intermediate and tall tree strata (D+E). Field and simulation results show that GeoCBI is more consistently related to spectral reflectance than CBI for different ranges of burn severities, while keeping its ecological meaning.

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1. Introduction

Forest fires can be a major ecological disturbance agent that modifies landscapes, especially when normal fire frequencies and/or intensities are modified. The main negative fire effects are vegetation biomass loss, soil degradation (Doerr et al., 2006; García-Haro et al., 2001; Lewis et al., 2006; Salgado et al., 2004) and greenhouse gas emissions (Andreae & Merlet, 2001; Nagahama & Suzuki, 2007; Narayan et al., 2007). Due to the wide range of spatial and temporal scales, the interpretation of causal factors, fire effects and ecosystem responses are a growing challenge for both researchers and managers (Lentile et al., 2006).

To clarify the complex interaction between fires and ecosystems, two different orders of fire effects have been proposed (Key, 2006): the *first- and the second order effects*. First-order effects are caused directly by combustion, second-order effects are caused by agents other than fire, with those agents being created or established as a result of fire. First order effects tend to occur in the short-term after the fire, while the second-order effects occur mainly in a longer-term. Therefore, this temporal dimension could be expressed more appropriately using the terms short-term and long-term fire effects. Both *first* and *second-order* effects on vegetation and soil can be estimated in terms of “fire or burn severity” (Chuvieco et al., 2006; De Santis & Chuvieco, 2007;

Jain, 2004a; Key & Benson, 2005; Lentile et al., 2006; van Wagtendonk et al., 2004; White et al., 1996). For this study the term “burn severity” will be used to account for the amount of changes on a burned area with respect to the pre-fire conditions (Key & Benson, 2005).

In the short-term after the fire, a detailed and rapid knowledge of the level of damage and its spatial distribution (burn severity map) is essential to quantify the impact of fires on landscapes (van Wagtendonk et al., 2004), select and prioritize treatments applied on site (Bobbe et al., 2001; Patterson & Yool, 1998), plan and monitor restoration and recovery activities and, finally, to provide baseline information for future monitoring (Brewer et al., 2005).

Several methods have been proposed to estimate burn severity from field assessments of post-fire soil and vegetation conditions, considering a wide range of variables (Table 1). The definition of a common field index is critical to quantify fire effects, since it would ensure consistent and comparable results (Key, 2006). A relevant attempt in this direction is the Composite Burn Index (CBI, Key & Benson, 2005), which was developed as an operational methodology for burn severity assessment on a national scale in the U.S, in the framework of the FIREMON (Fire Effects Monitoring and Inventory Protocol) project. The CBI was initially designed for long-term effects assessments, but it has been used also in short-term effects evaluation. This index is better adapted to estimate burn severity variations in forests than in shrubland or grassland (Key & Benson, 2005). CBI provides a balanced, continuous index of severity in ecological terms. Additionally, it was designed to be operationally retrieved from medium-resolution remotely sensed data, namely Landsat TM.

* Corresponding author.

E-mail addresses: angela.desantis@uah.es (A. De Santis), emilio.chuvieco@uah.es (E. Chuvieco).

Table 1
Field variables assessed to estimate burn severity

Variables assessed in the field	Reference
Percentage of tree basal area mortality	(Chappell & Agee, 1996)
Decrease in plant cover	(Jain & Graham, 2004b; Rogan & Yool, 2001)
Volatilization or transformation of soil components to soluble mineral forms	(Turner et al., 1994; Wang, 2002; Wells & Campbell, 1979)
Proportion of fine branches remaining on the canopy	(Moreno & Oechel, 1989)
Degree of canopy consumption and mortality	(Doerr et al., 2006; Kokaly et al., 2007; Kushla & Ripple, 1998; Patterson & Yool, 1998; Rogan & Franklin, 2001; Ryan & Noste, 1985)
Char and ash cover	(Smith et al., 2005b)
Composite Burn Index (CBI, Key & Benson, 2005) and its modifications	(Chuvieco et al., 2007; Cocke et al., 2005; De Santis & Chuvieco, 2007; Epting et al., 2005; Key & Benson, 2005; Miller & Yool, 2002; Miller & Thode, 2007; Sorbel & Allen, 2005; van Wagtenonk et al., 2004; Wimberly & Reilly, 2006)

In the field protocol, average post-fire conditions of soil and plant communities are visually evaluated in 30 m-diameter field plots. The CBI field form considers five vertical strata, organized in a hierarchical structure (Table 2), and mainly takes into account litter and fuel consumption, changes in soil colour, foliage or cover alteration, canopy mortality and char height. These attributes are rated in numerical scores ranging from 0 (unburned) to 3 (completely burned), by at least two field observers. Different attributes per stratum are scored and averaged into understory, overstory and overall composite rating.

Numerous recent studies have attempted to estimate CBI from remotely sensed images using empirical models. The most common approach has been to correlate CBI values with spectral indices, such as the Normalized Burn Ratio (NBR, Key & Benson, 1999), the difference between pre- and post-fire NBR (dNBR, Key & Benson, 1999) and, more recently, the relative delta Normalized Burn Ratio (RdNBR, Miller & Thode, 2007):

$$\text{NBR} = \frac{\rho_4 - \rho_7}{\rho_4 + \rho_7} \quad (1)$$

$$\text{dNBR} = \text{NBR}_{\text{PRE-FIRE}} - \text{NBR}_{\text{POST-FIRE}} \quad (2)$$

$$\text{RdNBR} = \frac{\text{NBR}_{\text{PRE-FIRE}} - \text{NBR}_{\text{POST-FIRE}}}{\sqrt{\text{ABS}(\text{NBR}_{\text{PRE-FIRE}}/1000)}} \quad (3)$$

where ρ_4 and ρ_7 are the reflectance of band 4 (near infrared, NIR) and band 7 (short wave infrared, SWIR) of Landsat TM, respectively.

The results are very diverse, with determination coefficients ranging from 0.01 to 0.81 (Kasischke et al., 2008) although most studies have shown medium to high values ($R^2 > 0.55$). The lowest correlations were generally explained by the non-linear relationship between CBI and dNBR, together with a signal saturation when CBI > 2.3 (van Wagtenonk et al., 2004). Moreover, most studies do not consider the Root Mean Square Error (RMSE) between observed and estimated values or the presence of systematic biases (deviation from the 1:1 linear regression).

These inconsistencies found in the CBI-NBR relations may be caused by two factors. Either the CBI cannot properly be retrieved from reflectance measurements, or the NBR is not sensitive enough to CBI variations. There is not a consistent study in the literature which

clarifies this point. Some papers have criticized the use of CBI as a good measure of burn severity, especially in boreal ecosystems (French et al., 2008). Instead, others have shown that the NBR may not properly represent transitions between pre and post-fire reflectance (Roy et al., 2006).

The main purpose of this paper is to test the potential problems in retrieving CBI from reflectance measurements in the optical domain (0.4 to 2.5 μm). The main components of CBI affecting canopy and plot reflectance will be considered by using a simulation approach. As a result of this analysis, a new version of the CBI will be proposed, which should be more consistently retrieved from remotely sensed data. This new index should preserve both the ecological meaning and the straightforward approach of CBI. The differences between the original and new CBI will be presented using three different Mediterranean study areas in Spain and Portugal. Both the simulation and field data will be restricted to the initial assessment conditions. Since previous studies have shown the potentials of other retrieving approaches (Chuvieco et al., 2006; De Santis & Chuvieco, 2007; De Santis et al., 2009), this paper will not discuss the adequacy of the NBR as a predictor of CBI. However, the changes in NBR as a result of CBI factors controlling reflectance variations will be considered.

1.1. The effect of vegetation coverage on the spectral response: a simulation analysis

In the computation of the average CBI of the total plot, all vegetation strata are assigned the same weight, regardless of their degree of coverage within the field plot. However, from a remote sensing point of view, the spectral response of the total plot is strongly related to the vegetation coverage per stratum, which is not commonly used to compute the CBI. As a result, the same CBI value could be obtained from different vegetation cover fractions, which would have different reflectance. Consequently, a CBI value does not have a unique spectral signature, and therefore estimations based on remotely sensed data will inevitably produce errors, regardless the specific spectral technique applied (spectral indices, classification methods, etc).

The influence of vegetation cover on reflectance of different strata within a given plot can be described using two variables: Leaf Area Index (LAI), defined as the area of leaf surface per unit of soil surface (Ceccato et al., 2002a), and Fraction of Cover (FCOV), characterized as the percentage of vegetation coverage with respect to the total plot area. Within the CBI, the change in LAI caused by the fire is indirectly considered in the shrubs and small trees layer (named “percentage change in cover” in stratum C), and in the upper tree layer (D+E) as a percentage change of crown foliage volume (for green, scorched and torched leaves).

The fraction of vegetation cover per stratum within each plot (FCOV) is not considered in the CBI calculations, but it has an important effect on plot reflectance. In order to quantify this effect, a simulation analysis was carried out assuming a tree-covered plot. Two widely used radiative transfer models were linked for the simulation: PROSPECT (at leaf level, Jacquemoud, 1990) and GeoSail (at canopy level, Huemmrich, 2001).

At leaf level, the CBI field form records the percentage of two types of leaves which are present on the canopy: green (undamaged) and brown (damaged) leaves. PROSPECT was then run in forward mode to

Table 2
Hierarchical structure of the CBI

CBI of total plot	Understory	A: Substratum B: Herbs, low shrubs and tree <1 m C: Tall shrubs and trees =1 to 5 m
	Overstory	D: Intermediate trees =5 to 20 m E: Large trees >20 m

Table 3
Input values of simulation at leaf level

Inputs	Green leaf	Brown leaf
Leaf structural parameter: N	2.5	2.5
Chlorophyll a+b content: Ca+b ($\mu\text{g}/\text{cm}^2$)	70	20
Equivalent water thickness: Cw (g/cm^2)	0.048	0.0008
Dry matter content: Cm (g/cm^2)	0.035	0.035
Brown pigments content: Cs (%SLW)	0.2	1.5

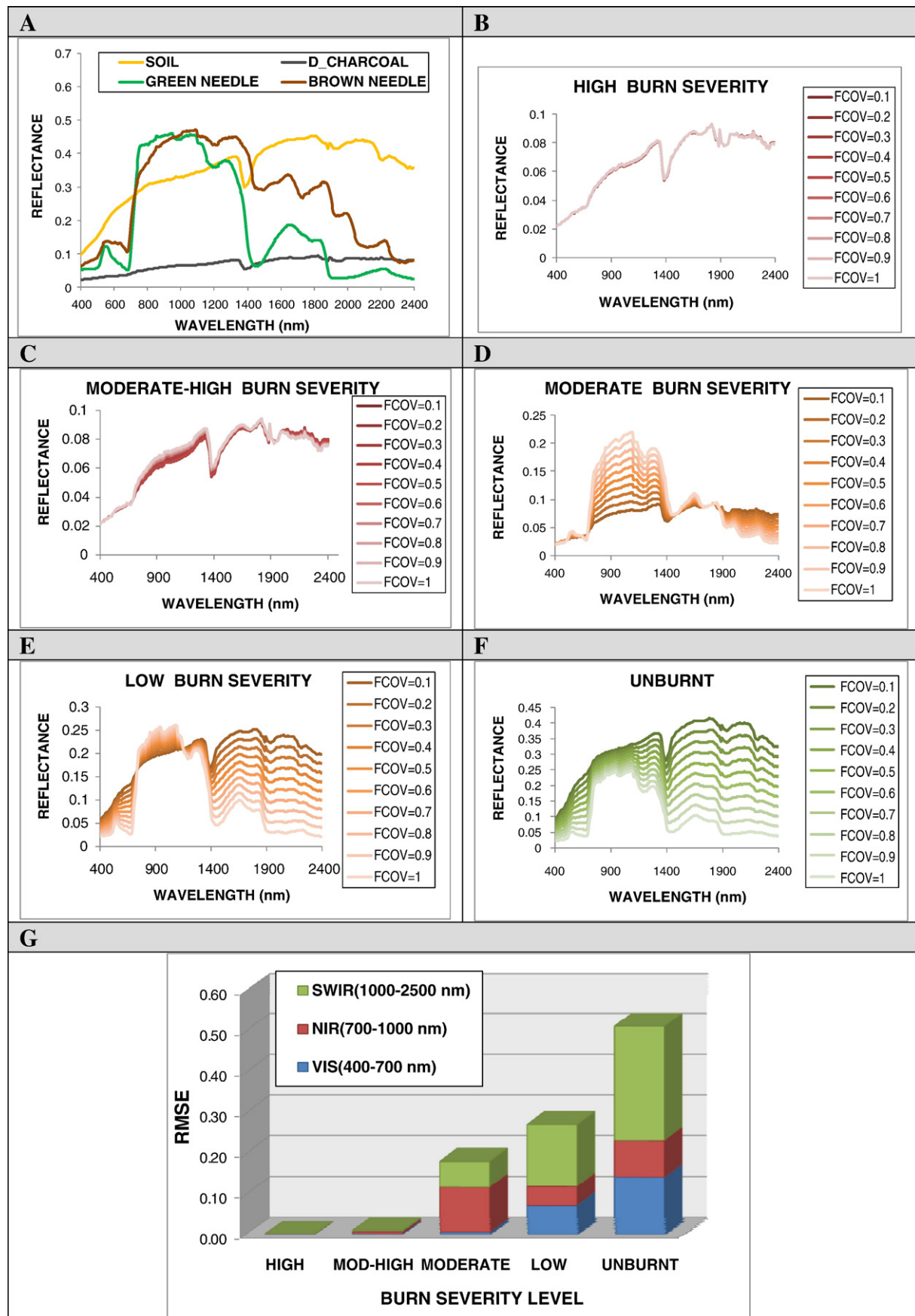


Fig. 1. (A) Input spectra of Geosail ; (B–F) simulation results and (G) RMSE between spectra for each burn severity scenario.

simulate only one representative spectrum for green and brown leaves. The model input values (Table 3) were fixed using both literature and field data (De Santis et al., 2009).

The outputs of the PROSPECT model (i.e. the two simulated leaf spectra, Fig. 1, A) were input to the GeoSail model to simulate five burn severity scenarios (from high severity to unburnt) using parameters based on our own field experience (Table 4). For each severity scenario, 10 spectra were generated by keeping fixed all inputs and varying only the FCOV from 0.1 to 1. Both the understory and the overstory were assigned the same FCOV value, which increased simultaneously in every simulation step.

Simulation results (Fig. 1, B–F) show that changes in FCOV have a higher effect as the burn severity level decreases, since the contribution of green vegetation is also higher. The RMSE computed between the cases with FCOV=0.1 (minimum) and FCOV=1 (maximum) (Fig. 1, G) shows that:

- ✓ in high and moderate-high burn severity scenarios, FCOV does not cause considerable changes in reflectance. Since post-fire LAI in vegetation strata are very low, it is the soil substratum which mostly contributes to the total reflectance.
- ✓ in the moderate burn severity scenario, FCOV affects principally the NIR and the SWIR, where changes in reflectance are in the range of 6% to 11%. The presence of green leaves on the top of the tree crowns modifies the total plot spectral response as FCOV increases.
- ✓ in low burn severity and unburnt scenarios, relatively small variations in FCOV have significant effects on NIR and SWIR reflectances (changes from 5% to 28%). Therefore, NBR is unlikely to produce accurate results unless the FCOV is taken into account. In these cases, the effect of FCOV is most evident, due to the higher LAI of the vegetation strata and the different substratum types.

1.2. New field index definition

Taking into account the simulation analysis results and field experience, a series of modifications in the original CBI are proposed to improve in the retrieval of burn severity from remotely sensed images, maintaining a similar ecological meaning to the original CBI. The CBI estimates the percentage of change with respect to the pre-fire scenario, but does not include information about its spatial extent.

Therefore, new variables per stratum were added to the original field form (in grey boxes in the Fig. 2):

1. The percentage of changes in the LAI. In accordance with the original CBI variables, this variable was scored from 0 to 3 and was estimated for strata C, D and E (Table 1), and was added as a rating factor. LAI replaced the original “percentage change in cover” in stratum C, and the percentage change of crown foliage volume of layer D and E.
2. The fraction of coverage (FCOV) of the vegetation, with respect to the total plot for strata B, C, D and E (Table 1). FCOV was scored from 0 to 1, since it was conceived as a weighting factor. Consequently, the new version of CBI proposed, called GeoCBI (which is short for *Geometrically structured Composite Burn Index*), was computed as follows:

$$\text{GeoCBI} = \frac{\sum_{m=1}^{m_n} (\text{CBI}_m * \text{FCOV}_m)}{\sum_{m=1}^{m_n} \text{FCOV}_m} \quad (4)$$

where m refers to each vegetation stratum and n is the number of strata.

3. Finally, the amount of new sprouts was added to the original CBI rating factors since they can significantly modify background reflectance. New sprouts were included in the stratum B (herbs, shrubs and trees less than 1 meter: Fig. 2), because within few weeks after fire their size falls into that stratum.

These changes in the field protocol are quite straightforward and should hardly increase the time spent on field work.

1.3. Examples of field application of both indices

The comparison between both indices was carried out using post-fire field values and satellite data acquired from three different forest fires in Mediterranean ecosystems.

The main area (site 1, located in central Spain, Fig. 3 and Table 5) correspond to large forest fire that burned 13,000 ha in 2005, mainly covered by pine forest (*Pinus pinaster*) mixed with semi-deciduous oaks (*Quercus pyrenaica* and *Quercus faginea*).

Table 4

Parameterization of the model at canopy level (DCH= dark charcoal, strata B+C= understory, strata D+E= overstory, LAD= Leaf angle distribution; Rv to Rh= crown height to width ratio)

Simulation at canopy level						
Burn severity scenario		High	Moderate-high	Moderate	Low	Unburnt
CBI		3	2.85	2.45	0.9	0
Example						
Strata	Leaf type	–	–	Brown	Brown	Green
B+C	LAI	–	–	1.7	0.5	1
Strata	Leaf type	Brown	Brown	Green	Green	Green
D+E	LAI	0.01	0.1	0.7	1.8	2.5
Sub-stratum type		DCH	DCH	DCH	50% SOIL 50% DCH	SOIL
Other inputs (fixed)		LAD = spherical; Rv to Rh = 2.36; Sun zenith angle = 30°; Crown shape = cone; note: all simulations were performed from 400 to 2400 nm, at 10 nm-intervals.				

STRATA	BURN SEVERITY SCALE							FACTOR SCORES
	No Effect	Low		Moderate		High		
	0	0.5	1.0	1.5	2.0	2.5	3.0	
A SUBSTRATES								
%DEAD LEAVES ON THE SOIL=				SOIL DEPTH (cm)=				
Litter/Light Fuel Consumed	Unchanged	--	50% litter	--	100% litter	>80% light fuel	98% light fuel	
Duff	Unchanged	--	light char	--	50% loss deep char	--	Consumed	
Medium /heavy Fuel	Unchanged	--	20% consumed	--	40% consumed	--	>60% loss deep char	
Soil & Rock Cover/Color	Unchanged	--	10% change	--	40% change	--	>80% change	
A. Σ =		N=		$\bar{X}$ =				
B HERBS, LOW SHRUBS AND TREES LESS THAN 1 METER								
DOMINANT VEGETATION TYPE =				FCOV=				
%Foliage altered (blk- brn)	Unchanged	--	30%	--	80%	95%	100%+branch loss	
Frequency % Living	100%	--	90%	--	50%	<20%	0%	
New sprouts	Abundant	--	moderate-high	--	moderate	--	low-none	
B. Σ =		N=		$\bar{X}$ =				
C TALL SHRUBS AND TREES 1 TO 5 METERS								
DOMINANT VEGETATION TYPE =				FCOV=				
%Foliage altered (blk- brn)	0%	--	20%	--	60-90%	>95%	significant branch loss	
Frequency % Living	100%	--	90%	--	30%	<15%	<1%	
LAI change %	Unchanged	--	15%	--	70%	90%	100%	
C. Σ =		N=		$\bar{X}$ =				
D INTERMEDIATE TREES 5 TO 20 METERS								
DOMINANT VEGETATION TYPE =				FCOV=				
% Green (unaltered)	100%	--	80%	--	40%	<10%	none	
%Black/ Brown	0%	--	20%	--	60-90%	>95%	significant branch loss	
Frequency % Living	100%	--	90%	--	30%	<15%	<1%	
LAI change %	Unchanged	--	15%	--	70%	90%	100%	
Char Height	none	--	1,5 m	--	2,8 m	--	>5 m	
D. Σ =		N=		$\bar{X}$ =				
E BIG TREES >20 METERS								
DOMINANT VEGETATION TYPE =				FCOV=				
% Green (unaltered)	100%	--	95%	--	50%	<10%	none	
%Black/ Brown	0%	--	20%	--	60-90%	>95%	significant branch loss	
Frequency % Living	100%	--	90%	--	30%	<15%	<1%	
LAI change %	Unchanged	--	15%	--	70%	90%	100%	
Char Height	none	--	1,8 m	--	4 m	--	>7 m	
E. Σ =		N=		$\bar{X}$ =				

Fig. 2. The new version of the field form proposed for estimating burn severity as GeoCBI. New variables are in grey boxes.

The second area is composed of a small cluster of several fires, which occurred in 2006 in the north-western region of Spain (Galicia) (site 2, Fig. 3 and Table 5). Unlike the other two sites, in this area the main tree species is eucalyptus (*Eucalyptus globulus*) followed by pine trees (*Pinus pinaster*) and oaks (*Quercus robur*) with a thick understory of gorse (*Ulex europaeus*) shrubs.

Finally, the third area was located in north-eastern Portugal (site 3, Fig. 3 and Table 5). In this case, the forest fire burned a total of 568 ha, mainly composed of pine trees (*Pinus pinaster*) mixed with semi-deciduous oaks and Mediterranean shrubs.

Burn severity was assessed in the field in terms of both CBI and GeoCBI. A total of 128 field plots (Table 5) were selected within areas with homogeneous burn severity levels, as recommended by Key and Benson (2005), and gentle slopes. In each field plot both the original and the new field forms were filled in, digital photographs were taken

and the centre coordinates of the plot were recorded using a GARMIN 12 GPS (5–10 m of error in x and y).

In addition, spectral signatures corresponding to the field plots of site #1 were extracted from a post-fire Landsat 5 Thematic Mapper (TM) (acquired on August 5th, 2005; Table 5). Radiometric, geometric and illumination corrections were applied as in De Santis and Chuvieco (2007).

1.4. Performance of GeoCBI versus CBI

Fig. 4 (left) shows a comparison between both CBI and GeoCBI field values. In site #1, the highest differences between both indices were observed between ~1.5 and ~2.6 severity values. In this range, CBI values generally tend to be lower than GeoCBI, especially when the FCOV of the tree stratum is under 50%, as in plot C (Fig. 4). In this plot

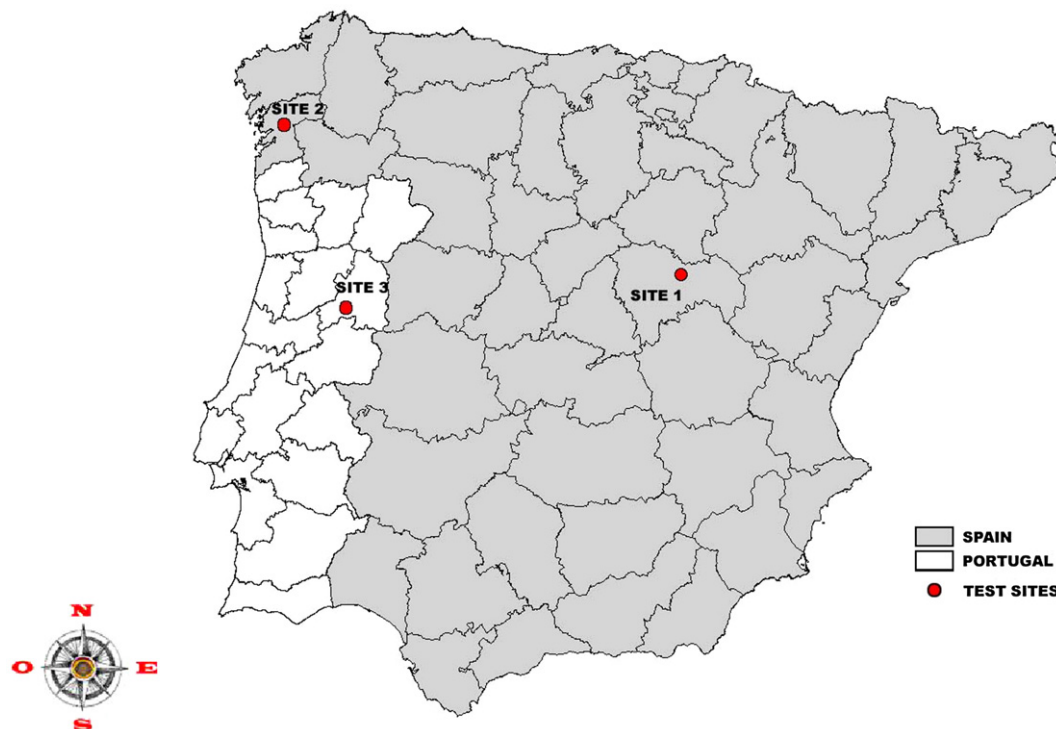


Fig. 3. Test site locations.

(see also case C, Fig. 5) the understory is highly affected, whereas the tree stratum still maintains green canopies despite its very low (10%) FCOV.

Exceptionally, in two of the plots CBI values are higher than GeoCBI (cases A and B, Figs. 4 and 5). In these two plots, the understory has a very high burn severity whereas its FCOV is very low (5–15%).

According to the results obtained in the site #1, in the site #2, CBI values are lower than GeoCBI, especially in correspondence of plots D and E (Fig. 5). These two plots present the same strata combination of the case C (understory highly affected, while the overstory has a very low severity value and low FCOV). In the site #3, there is a higher concordance between CBI and GeoCBI, except for plot F.

1.5. GeoCBI and CBI versus NBR

Although the main goal of this paper was not the analysis of NBR and CBI, but rather between CBI and reflectance, correlations of the proposed GeoCBI with NBR have also been included, since the NBR has been extensively used to retrieve CBI values from satellite data (Lentile et al., 2006; Kasischke et al., 2008). Results for site #1 (Fig. 4, right), show that NBR has a significantly better fit with GeoCBI than with CBI ($R^2=0.54$ and $R^2=0.32$, respectively), and a lower dispersion of the values from the linear trend. The same pattern is observed in the site #2. In this case the difference in determination coefficient between both indices is lower (GeoCBI $R^2=0.66$; CBI $R^2=0.61$) because the burn severity estimation obtained using both indices is very similar (see Fig. 4, left, site #2). Finally, in site 3, since the burn severity

values of both indices are almost the same, the differences in R^2 are not significant (GeoCBI $R^2=0.41$; CBI $R^2=0.43$).

1.6. Relationship between both field indices and satellite data

The relationships between satellite data and GeoCBI were found more consistent than for CBI. Since FCOV has an important effect on plot reflectance, the same CBI with different FCOV values generates different reflectances, as it is shown in Fig. 6. Plots P95, P86, and P23 have the same CBI, in spite of having very different structures, compositions and burn severity. However, they have different GeoCBI values, as well as different spectral reflectances from the Landsat TM image. In contrast, the second example of Fig. 6 shows another three field plots (P7, P102, P21), having the same GeoCBI, composition structure and quite similar spectral signatures, but different CBI values. In short, CBI variations are influenced by the absence/presence of highly affected shrubs that have a very low FCOV, whereas GeoCBI takes into account this factor and thus correctly assigns the same value.

When using satellite data to estimate burn severity of plots with high severity values in the substratum and understory but not the upper canopy, the estimations would strongly differ when a low FCOV or a high FCOV are present. The former situation would provide a signal close to a char-scorched leaf, while the latter would resemble a green spectrum (Fig. 7A and B). The former would imply an over-estimation of burn severity, while the latter an under-estimation. Both situations would have different GeoCBI values.

Table 5

Test sites information: fire name and date, burnt area, field work and satellite data

Test site #	Fire name (province)	Fire date (dd/mm/yyyy)	Burnt area (ha)	Field work		Sensor	Image date (dd/mm/yyyy)
				Date (dd/mm/yyyy)	# plots		
1	Riba de Saelices (Guadalajara)	16/07/2005	12,892	From 18/08/2005 to 28/09/2005	104	Landsat 5 TM	05/08/2005
2	Serrapio (Pontevedra)	4/08/2006	3360	06/10/2006	10	SPOT 5 HRG	06/09/2006
3	Familação da Serra (Guarda)	09/07/2006	568	From 17 to 19/08/2006	14	Landsat 5 TM	05/09/2006

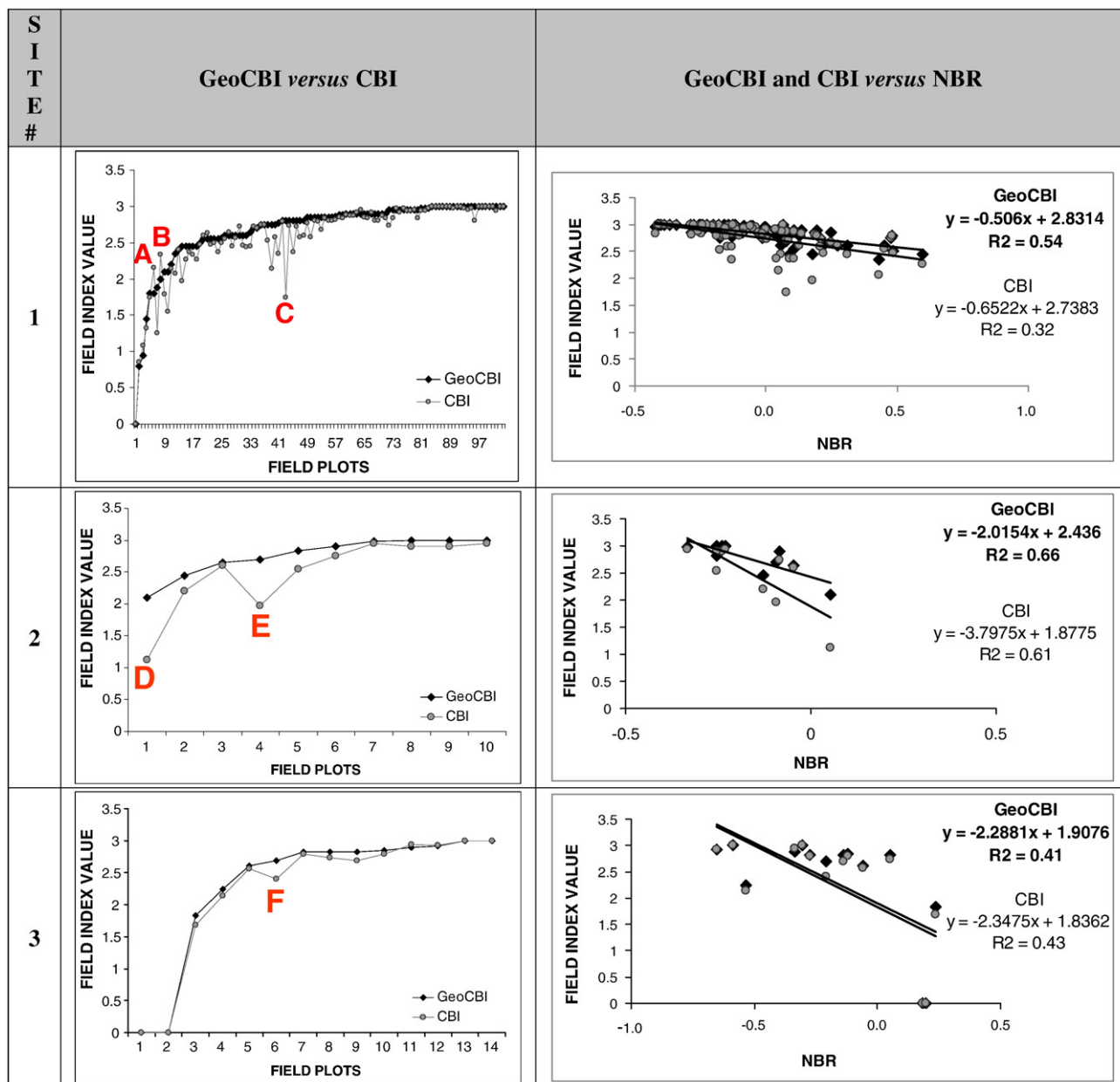


Fig. 4. Left: comparison between CBI and GeoCBI (field plots are ordered in terms of GeoCBI, from 0 to 3). Right: correlation between NBR and both field indices.

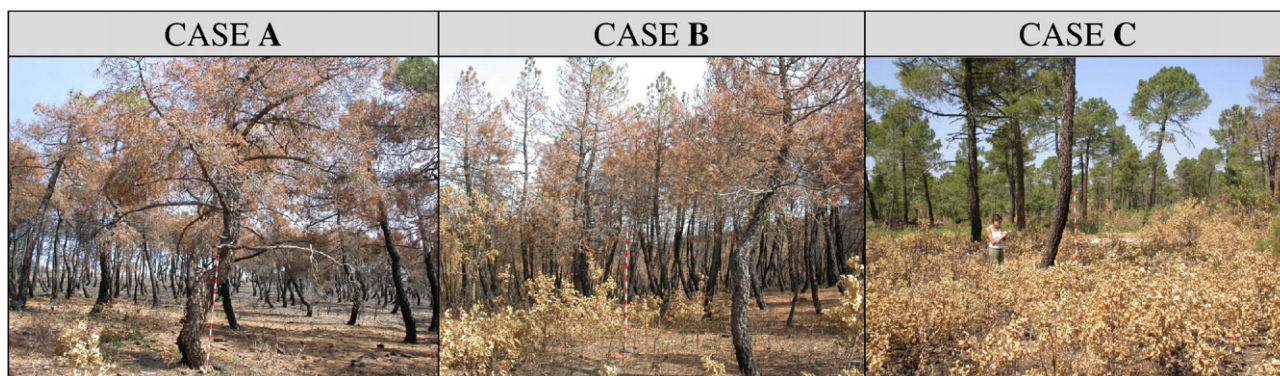


Fig. 5. Field plots of site #1 which have the highest overestimation (Case A and B) and underestimation (Case C) of burn severity using the original CBI with respect to GeoCBI.

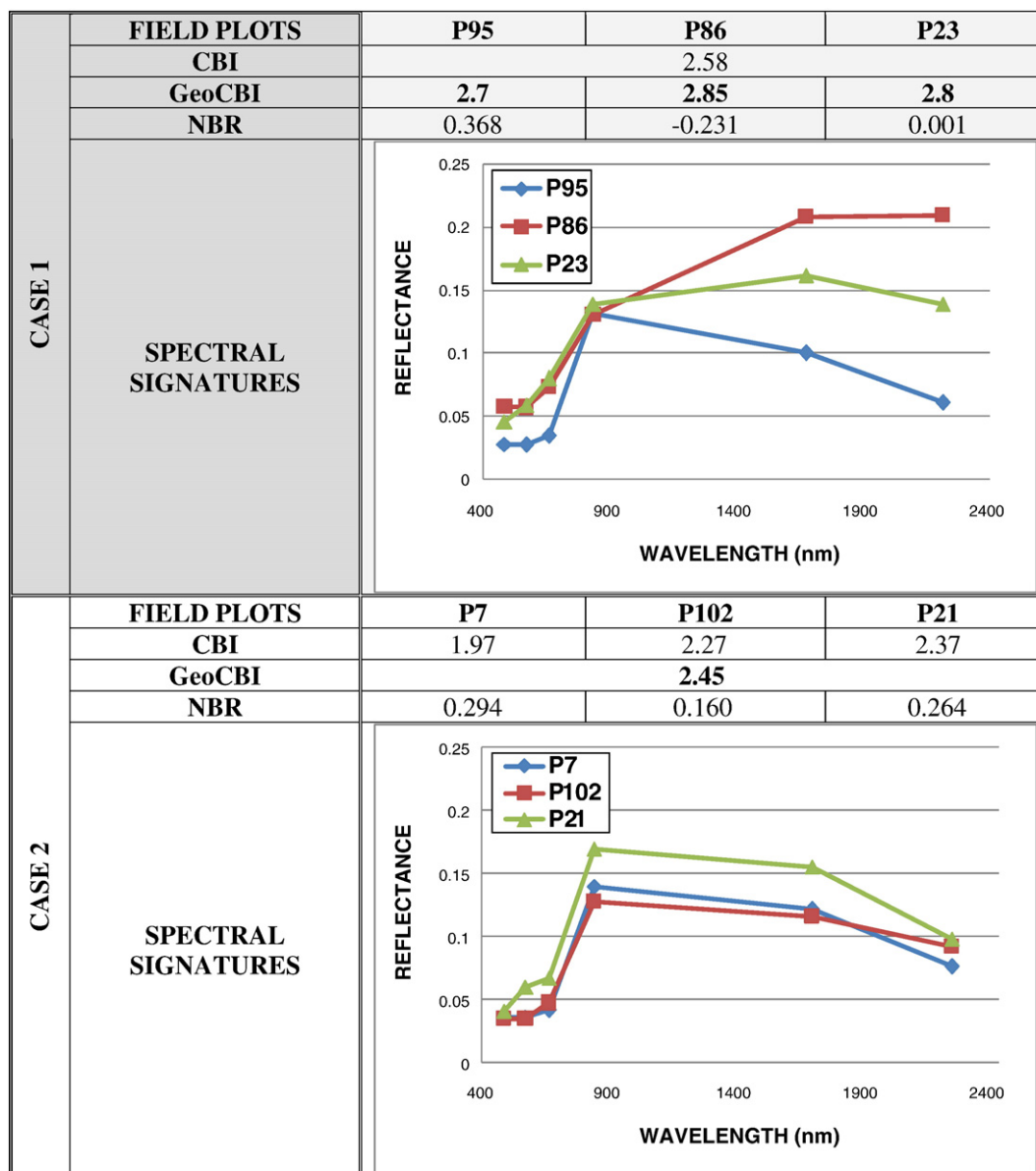


Fig. 6. Comparisons between CBI, GeoCBI, NBR and spectral signatures of the same plots of site #1.

However, the estimation of severity of the understory and substratum when the overstory has a high FCOV value (Fig. 7B and C) remains an open problem that GeoCBI cannot solve. In these cases, when the understory has a high burn severity, GeoCBI would underestimate the plot severity, similarly to the CBI. The limitation in this case is common to most optical remote sensing techniques, since satellite observed reflectance comes from the upper canopy when it is dense. Nevertheless, from the point of view of short-term post-fire management, such cases normally do not require a rapid

intervention, since they have mainly intermediate to low values of GeoCBI.

2. Conclusions

Burn severity is a key factor in post-fire management in order to quantify and map fire impacts and, consequently, to plan the mitigation and rehabilitation treatments in the burned areas. Post-fire field assessment of burn severity is costly in terms of time and economic



Fig. 7. Examples tree stratum compositions: A) overstory with green canopies, a low FCOV and a dark charcoal substratum; B) overstory with green canopies, a high FCOV and a dark charcoal substratum; C) overstory with green canopies, a high FCOV and an unburnt soil substratum.

resources and does not provide a good spatial coverage. However, this field work is a source of ground truth which is important to understand the critical variables that determinate burn severity and, subsequently, to calibrate and validate remote sensing methods.

Although CBI is the most widely used field index, the retrieval of this index from remotely sensed data has proved to be inconsistent in some ecosystems. The data used in this study shows that GeoCBI provides a better characterization of the field plots than the original CBI, as well as a higher correlation with the spectral signatures extracted from Landsat TM data, and consequently, a higher correlation with well known spectral indices (such as the NBR). By introducing the FCOV as a weighting factor, the spectral mixture of the different plot components/variables are better represented, without adding a significant effort to the field work. In addition, GeoCBI maintains the ecological meaning of CBI. For post-fire management purposes, the correction the of burn severity overestimation in the cases described can help to maximize resources by skipping burnt areas that are less critical.

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